ELAINE GAY E. SOBREMONTE

Computer Engineering Graduate | Embedded Systems | Electronics | Machine Learning

Calamba, Laguna | elainesobremonte@gmail.com | 0949 - 988 - 4195

PROFESSIONAL SUMMARY

Computer Engineering graduate with hands-on experience in embedded systems, electronics, machine learning, and computer vision. Experienced in integrating hardware and software solutions using Raspberry Pi, microcontrollers, sensors, and ML models, with project leadership experience in technical development.

EDUCATION

Bachelor of Science in Computer Engineering **2022 - 2026**
National University - Laguna

INTERNSHIP EXPERIENCE

Engtek Precision Philippines Inc. | Process Engineering Intern | April 2026 – May 2026

- Developed and presented engineering dashboards for breakdowns, downtime, cycle time, and capacity, enabling clearer visualization and monitoring of engineering data.
- Revised and updated Work Instructions for Machining and Visual Inspection and created new WIs from scratch, improving process documentation consistency and accuracy.
- Analyzed Excel-based engineering data and collaborated with personnel on documentation and reporting, supporting more organized process monitoring and decision-making.

TECHNICAL PROJECTS

SwineWatch: Surveillance and Alert System for Early Signs of African Swine Fever in Pigs Using Machine Learning in CALABARZON, Philippines | Project Manager | 2025 – 2026

- Led a team in developing a machine-learning-based monitoring and alert system using thermal imaging and computer vision, supporting early identification of potential ASF indicators in pigs.
- Coordinated Raspberry Pi 4, Pi Camera V3, MLX90640, and SIM800L integration, enabling automated monitoring and SMS alert capabilities.
- Oversaw YOLOv8 implementation for pig detection, behavior, and skin-condition classification, supporting automated analysis of monitoring data.

AI-Powered Mosquito Species Classifier | Project Manager | 2024

- Led a team in developing an AI system using recorded mosquito wingbeat signals, enabling automated species identification.
- Coordinated MATLAB and ResNet18 transfer learning implementation, enabling wingbeat-based species classification.
- Managed preprocessing, model training, evaluation, and GUI integration, resulting in an integrated wingbeat-to-species prediction system.

CERTIFICATIONS

- **Certified in Cybersecurity (CC)** — ISC2 | August 2026
- **National Certificate II in Computer Systems Servicing (NC II)** — TESDA | September 2025
- **Basic Occupational Safety and Health for Safety Officer 2 (BOSH SO2)** — DOLE | January 2026
- **Lean Six Sigma Yellow Belt Certification** — Prof. Dr. Marcelo Machado Fernandes | January 2026
- **Lean Six Sigma White Belt Certification** — Ask Lex PH Academy (ALPHA) | February 2025

TECHNICAL SKILLS

Programming: Python, C++, Java, HTML
Embedded & Electronics: Raspberry Pi, Arduino, ESP8266, PCB Design, Sensor Integration
Machine Learning: YOLOv8, PyTorch, TensorFlow, ResNet18, Random Forest, 1D-CNN
Computer Vision & Data: OpenCV, Roboflow, Data Preprocessing
Web & Tools: HTML, CSS, JavaScript, MATLAB, Flask, Firebase, Git/GitHub, VS Code, Google Colab